Comprehensive Results Report

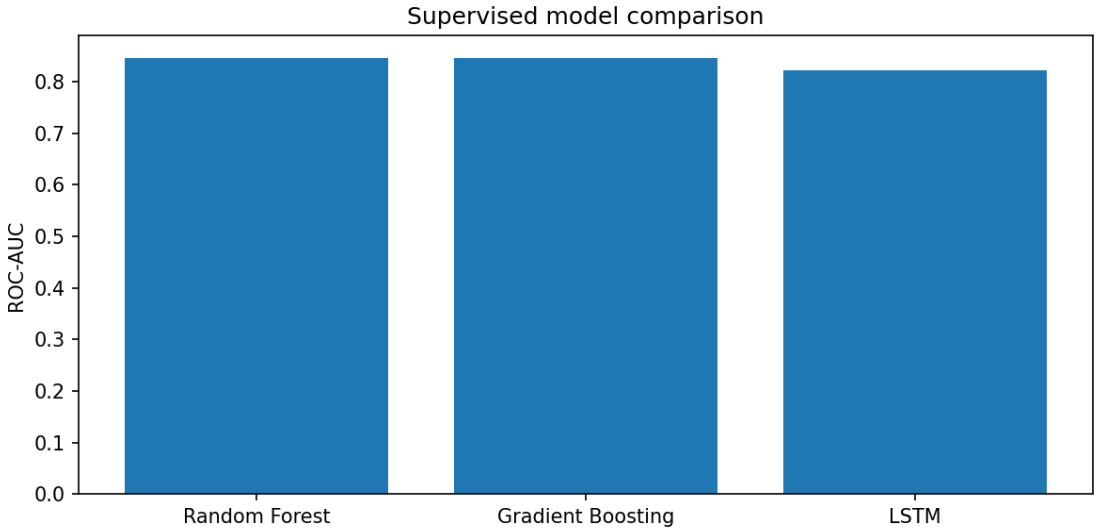
CHE 7700 Tennessee Eastman Process Analytics Project

ANN_ML on the Process control systems and Intelligent fault detection with Process Monitoring for Acetic Acid Production plant.

GitHub Repo link

In this machine learning research, I was able to demonstrate a strong array of (8) dimensional reduction with (4) – Clustering analysis. On an holistic scale, I was able to do a full technical analysis and deep dive report on results obtained.

Industrial framing: chemical process automation for acetic acid production
Modeling baseline: TEP00 = normal process benchmark; TEP01–TEP15 = faulty operation.



Summary view of binary supervised model ROC-AUC results.

Summary Report

This project report explains the results contained in the delivered CHE 4230 project package. The Tennessee Eastman Process (TEP) data were used as a surrogate industrial process dataset for fault monitoring, dimensionality reduction, clustering, and supervised classification. The central binary task was normal-versus-faulty detection, with *TEP00 treated as the normal benchmark* and TEP01–TEP15 *treated as faulty operations*.

After harmonizing the flowsheet to the common 50-sensor subset, the final dataset contained 7,286 observations from 16 files. The training set used 5,816 rows and the test set used 1,470 rows. The dataset is highly imbalanced: only 481 rows come from the normal benchmark while 6,805 rows correspond to fault cases.

The main conclusions are:

(1) the harmonized sensor set still preserves strong fault information.

(2) ordinary **PCA** (Principal component analysis) is not the most suitable in dimensional analysis.

(2) manifold-style embeddings on a balanced subset reveal more structure than linear reductions alone.

(3) **HDBSCAN** produced the strongest clustering alignment on the subset, especially for t-SNE and the UMAP-style fallback embedding.

(4) the supervised models all achieved similar binary detection performance, with Random Forest achieving the highest ROC-AUC of 0.846 (84.6%) and an accuracy of 0.934 (93.4%); and

(5) the LSTM converged quickly but did not materially outperform the tree models on this formulation.

Table 1. Core dataset summary.

metric	value
n_files (TEP00 – TEP15)	16
common_sensor_count	50
total_rows	7286
train_rows	5816
test_rows	1470
normal_rows	481
faulty_rows	6805

Project Scope and Data Preparation

The archived project follows the CHE 7700 brief but uses an expanded scope. The delivered package included EDA, multiple dimensionality-reduction and clustering combinations, SOM visualizations, and three supervised models: Random Forest, Gradient Boosting, and LSTM.

Reason is because the raw Excel files were not perfectly uniform, preprocessing standardized the analysis on the common 50-sensor subset. This was necessary because one workbook had fewer columns than the rest. The time ordering within each workbook was preserved, which made it possible to support dynamic methods such as DPCA, MPCA approximations, and sequence-based LSTM modeling.

Table 2. File-level audit after harmonization.

file	rows	used_cols
TEP00.xlsx	481	50
TEP01.xlsx	481	50
TEP02.xlsx	481	50
TEP03.xlsx	481	50
TEP04.xlsx	481	50
TEP05.xlsx	481	50
TEP06.xlsx	71	50
TEP07.xlsx	481	50
TEP08.xlsx	481	50
TEP09.xlsx	481	50
TEP10.xlsx	481	50
TEP11.xlsx	481	50
TEP12.xlsx	481	50
TEP13.xlsx	481	50
TEP14.xlsx	481	50
TEP15.xlsx	481	50

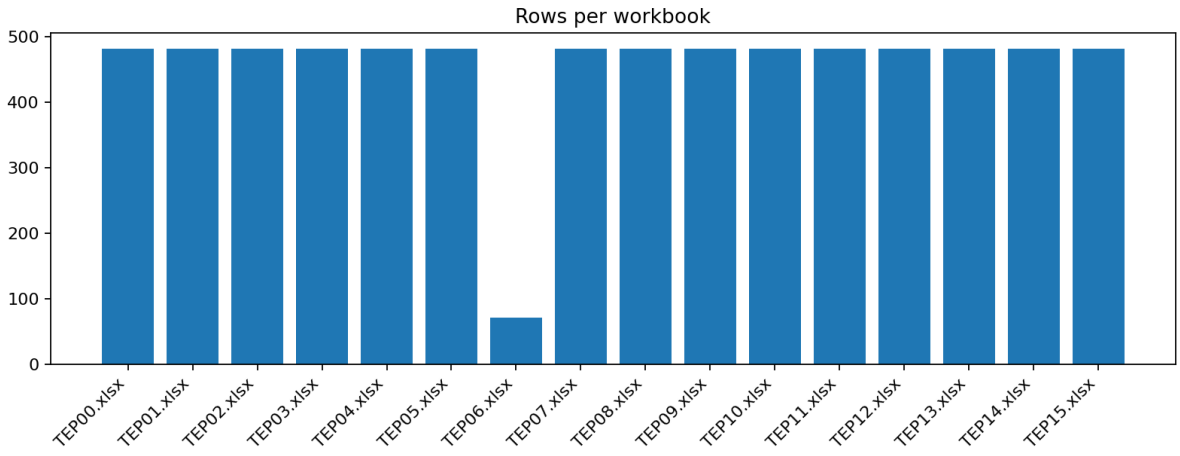


Figure 1. Sample counts by TEP file after harmonization.

Exploratory Data Analysis (EDA)

EDA was used to understand data quality, imbalance, sensor behavior, and broad fault signatures before modeling. The plot shows that nearly every file contains 481 rows except TEP06, which is much shorter. This matters because dynamic models can be sensitive to sequence length. The binary class balance is extremely skewed toward fault data, so accuracy alone is not sufficient for judging classifier quality.

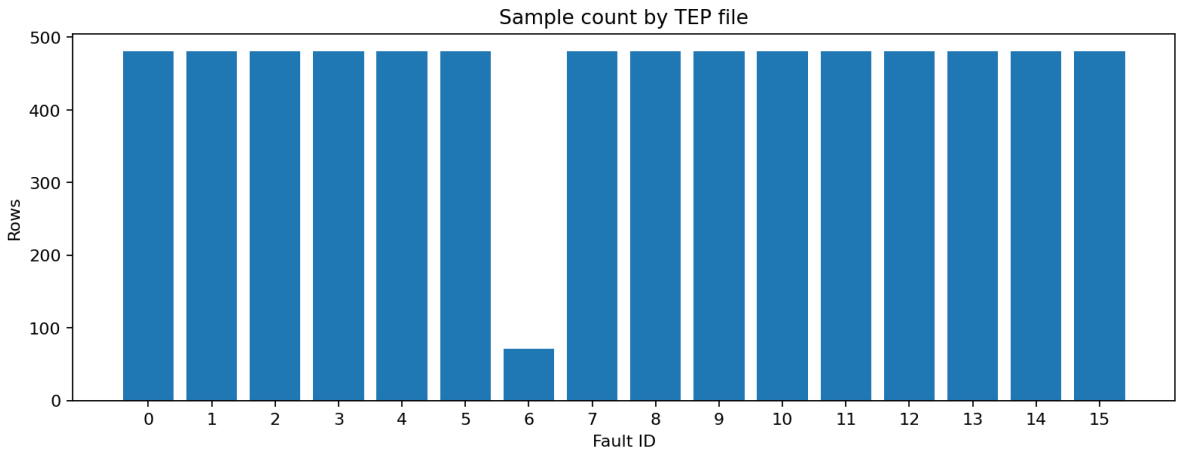


Figure 2. Class balance by fault ID. TEP00 is the normal benchmark; TEP01–TEP15 are faults.

The correlation heatmap indicates substantial dependence among sensors. This is expected in a coupled chemical process: flow, temperature, pressure, and composition variables do not evolve independently. Such correlation structure is one reason dimensionality reduction is appropriate. It also explains why tree ensembles can perform strongly even when individual variable importance is diffuse.

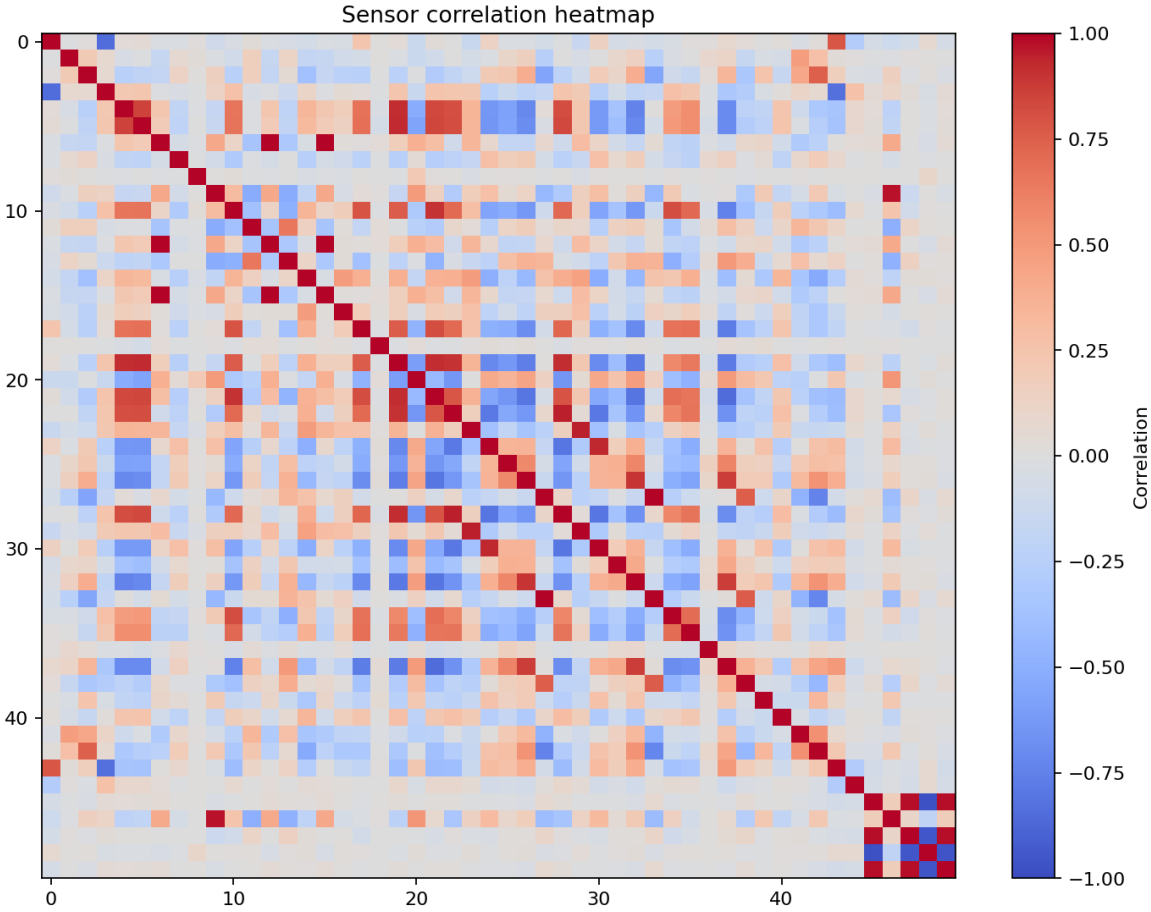


Figure 3. Correlation structure across the 50 common sensors.

The fault-pattern heatmap summarizes mean standardized sensor profiles by fault class. Several faults are visually distinct, while others remain close to the normal benchmark. This observation foreshadows the unsupervised results: some faults are cluster-separable, but many share overlapping process signatures in the common-sensor space.

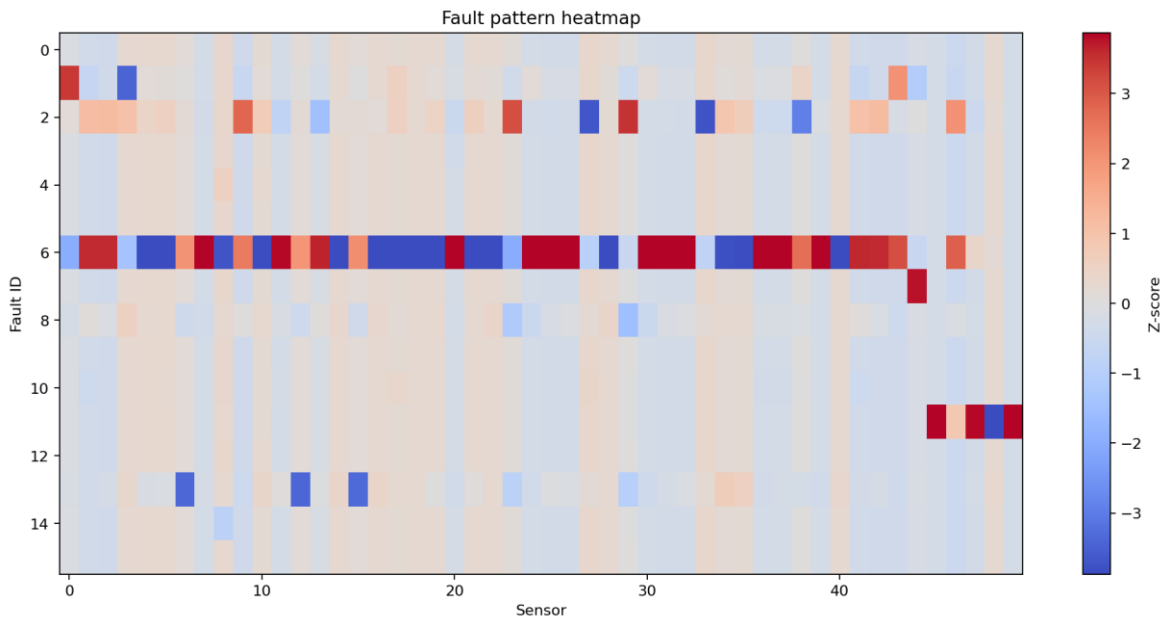


Figure 4. Fault mean profiles in standardized coordinates.

Normal-versus-faulty mean differences were largest for sensors x02, x43, x15, x06, x12, and x46. These variables therefore carry much of the broad separation between benchmark and faulty operations.

Table 3. Largest absolute mean shifts between normal and faulty operation.

sensor	delta	abs_delta
x02	6.4258	6.4258
x43	4.3339	4.3339
x15	-3.3542	3.3542
x06	-2.9649	2.9649
x12	-2.897	2.897
x46	2.8207	2.8207
x45	1.8309	1.8309
x01	1.1769	1.1769
x44	0.885	0.885
x49	0.8747	0.8747

The highest-variance sensors were x02, x15, x01, x06, x12, and x43. High variance alone does not prove fault relevance, but it does show which sensors carry the widest operating excursions across the archive.

Table 4. Highest-variance sensors across the harmonized dataset.

sensor	variance

x02	1,112
x15	429.0142
x01	384.1216
x06	327.2192
x12	311.0612
x43	221.6277
x46	166.3091
x45	41.7621
x14	27.2435
x19	19.4538

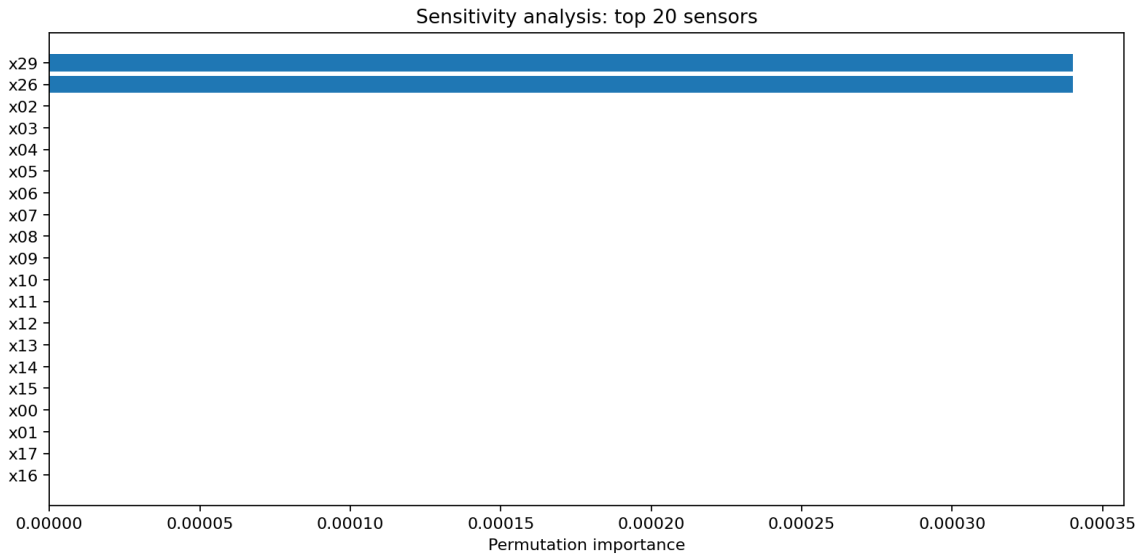


Figure 5. Permutation sensitivity ranking from the supervised fault-detection task.

The permutation sensitivity plot is relatively flat, with only small positive scores at the top. This suggests that fault detection is not dominated by a single sensor. Instead, the binary models appear to exploit distributed multivariate structure.

Dimensionality Reduction and Clustering Results

The unsupervised analysis was run on a balanced stratified subset rather than the full dataset. This design choice improves visual readability and prevents the severe normal/faulty imbalance from dominating the embeddings. Each embedding was clustered with K-means, HDBSCAN, and hierarchical clustering, and metrics were reported using ARI, NMI, silhouette score, and discovered cluster count.

A higher Adjusted Rand Index (ARI) means the discovered cluster assignments align more closely with the fault labels. Normalized Mutual Information (NMI) captures shared information between clustering and labels, while silhouette score measures geometric separation in the embedded space.

Table 5. Best unsupervised runs ranked by ARI on the subset.

ari	nmi	silhouette	n_clusters	embedding	clustering
0.2949	0.3878	0.8296	21	tsne	hdbscan
0.2948	0.4169	0.7758	16	umap_fallback	hdbscan
0.2395	0.347	0.7149	11	pacmap_fallback	hdbscan
0.0788	0.1974	0.6479	9	autoencoder	hdbscan
0.0521	0.21	0.6951	11	pca	hdbscan
0.044	0.1835	0.5453	9	trimap_fallback	hdbscan
-0.0063	0.0783	0.5957	7	dpca	hdbscan
-0.0116	0.0497	0.573	2	mpca	hdbscan
0.2182	0.379	0.675	16	tsne	hierarchical
0.1325	0.3785	0.5039	16	pacmap_fallback	hierarchical
0.1061	0.3322	0.6207	16	umap_fallback	hierarchical
0.0667	0.346	0.5544	16	trimap_fallback	hierarchical

The strongest overall unsupervised result in this archive was t-SNE combined with HDBSCAN (ARI = 0.295, NMI = 0.388, silhouette = 0.830). The next best result was the UMAP-style fallback embedding with HDBSCAN (ARI = 0.295, NMI = 0.417). More generally, HDBSCAN outperformed K-means and standard hierarchical clustering for most nonlinear embeddings.

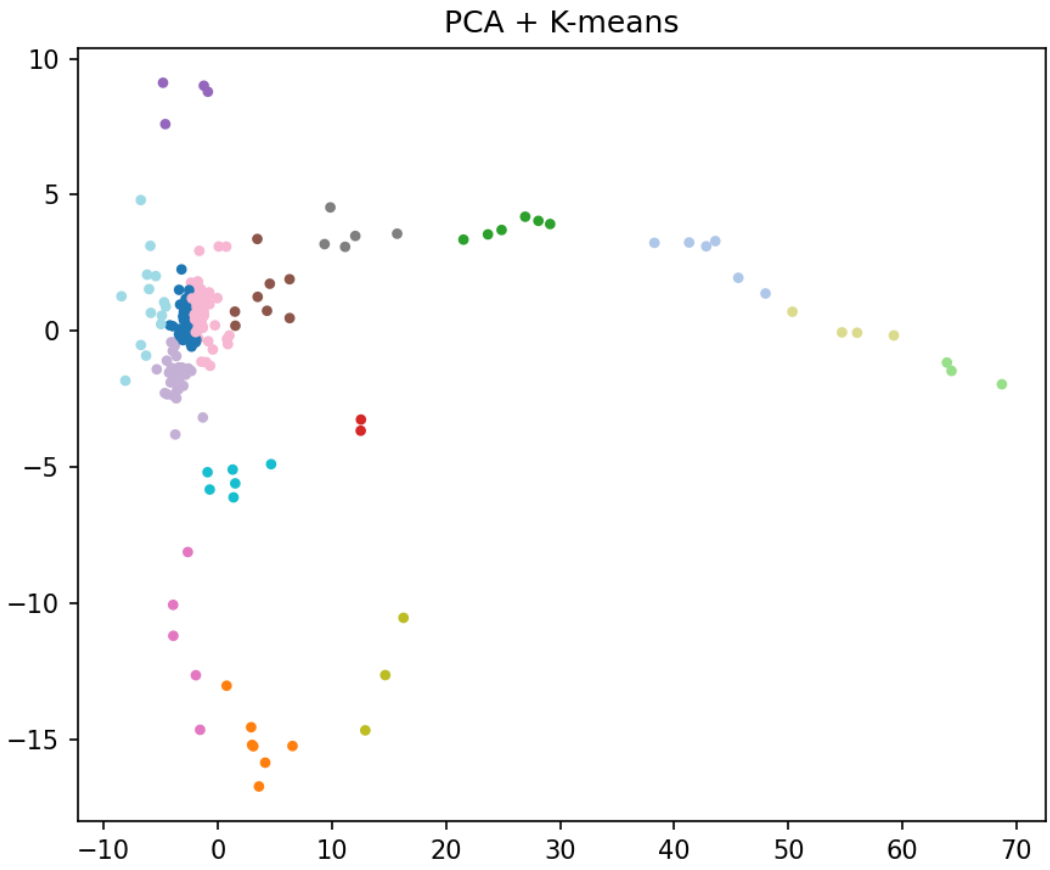


Figure 6. PCA embedding with K-means clustering on the balanced subset.

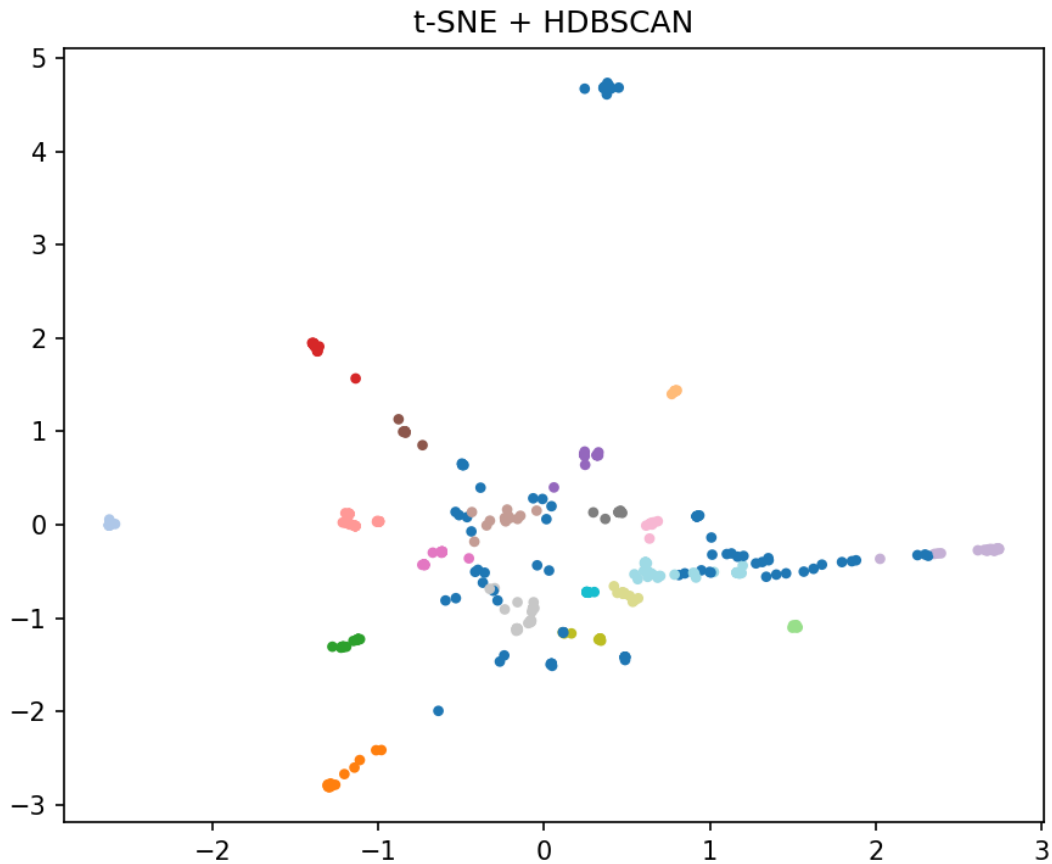


Figure 7. t-SNE embedding with HDBSCAN clustering on the balanced subset.

PCA gave interpretable but only modest clustering alignment. K-means on PCA produced a visibly structured map, but cluster correspondence to fault identity was weak (ARI ≈ 0.037). This indicates that the dominant linear variance directions do not cleanly separate the fault classes.

In contrast, t-SNE and the fallback manifold methods created embeddings in which local neighborhoods were more meaningful for clustering. HDBSCAN especially benefited from these embeddings because it can adapt to variable-density structure and does not force every point into a fixed number of clusters.

DPCA and MPCA were weaker than the best manifold methods in this archive. That outcome is plausible: the common-sensor harmonization and shortened TEP06 run likely reduced the value of the dynamic reconstructions, and the project used practical approximations rather than a full industrial-grade multiblock implementation.

The autoencoder embedding gave mid-tier performance. It was better than simple linear PCA in some pairings but did not outperform the best t-SNE/HDBSCAN or UMAP-style/HDBSCAN combinations.

Self-Organizing Map (SOM) Results

The SOM visualizations provide a complementary perspective on the subset structure. The U-matrix highlights distances between neighboring codebook vectors; darker regions suggest sharper topological boundaries. The hit map shows where observations accumulate on the map grid.

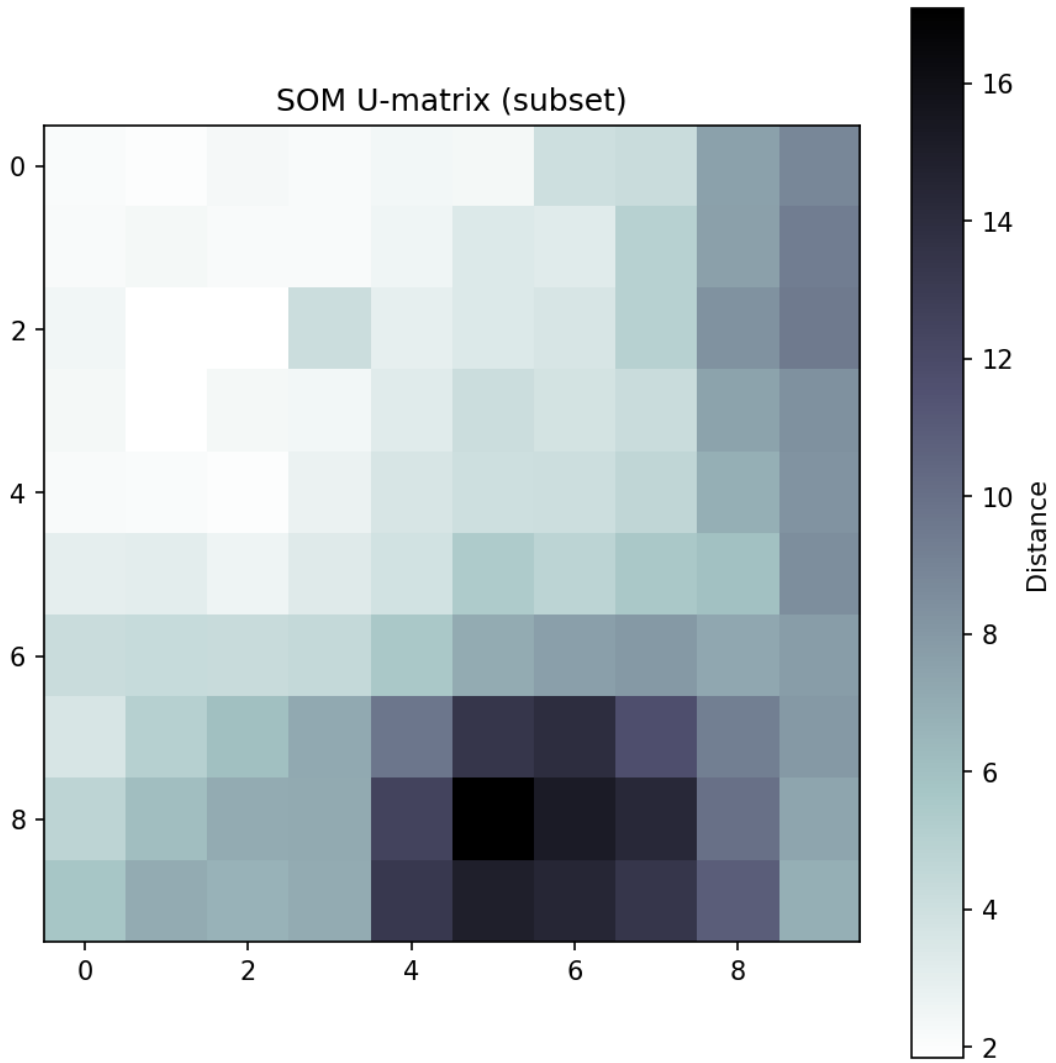


Figure 8. SOM U-matrix on the balanced subset.

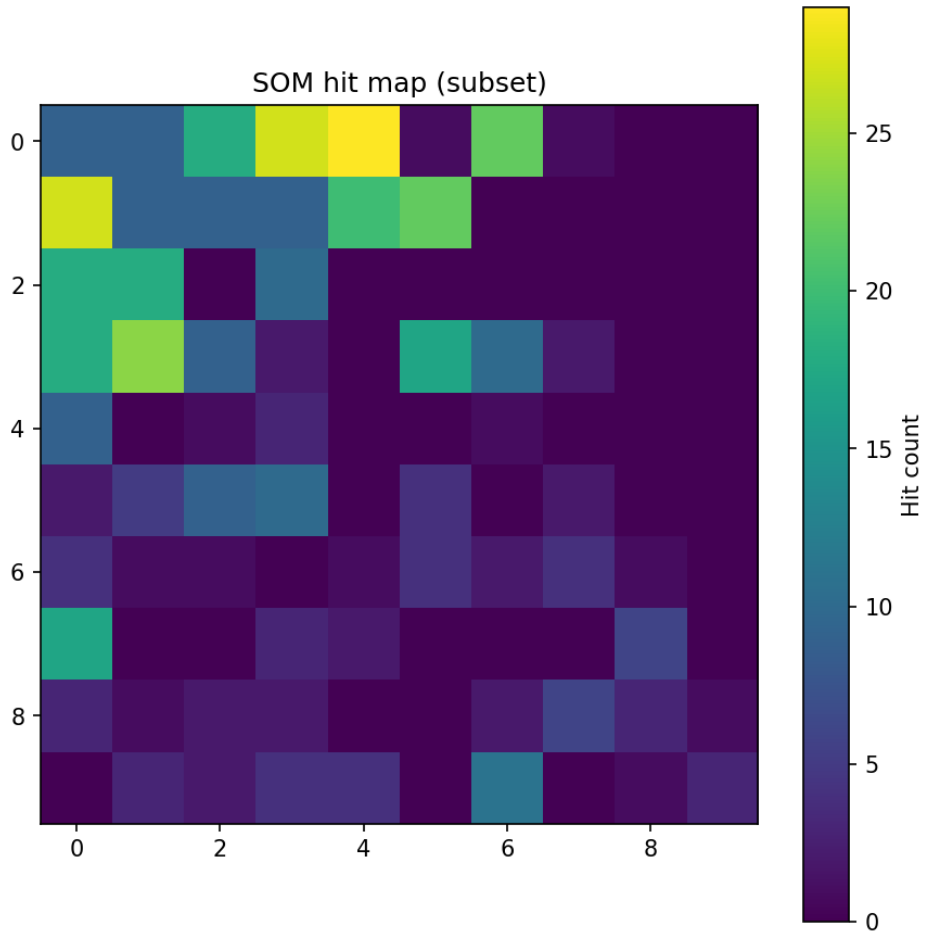


Figure 9. SOM hit map on the balanced subset.

Together, these plots suggest that the subset contains pockets of well-separated process behavior, but they are not uniformly distributed. That conclusion agrees with the clustering results: some faults form coherent neighborhoods, while others remain partially mixed based off of the intensity and profoundness of the coloration.

Supervised Fault Detection Results

Supervised modeling was carried out on the full harmonized dataset using the binary target normal-versus-faulty. This is a practical industrial framing: this plant usually needs to detect whether the current operating state is healthy or abnormal before attempting a more detailed diagnosis.

Table 6. Supervised model performance on the held-out test set.

model	accuracy	f1	roc_auc
Random Forest	0.934	0.9659	0.8465
Gradient Boosting	0.934	0.9659	0.8453
LSTM	0.9333	0.9655	0.8226

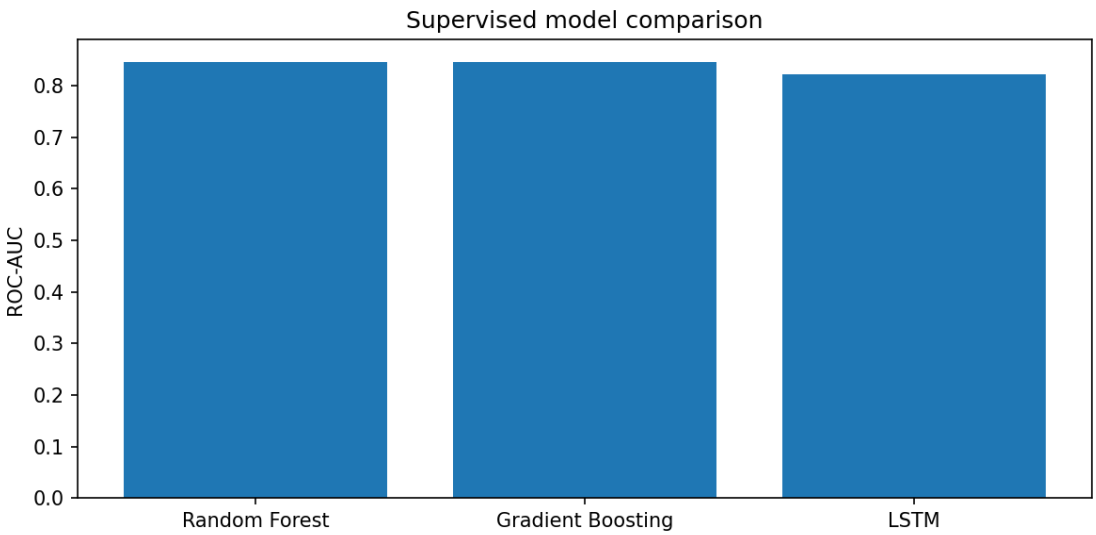


Figure 10. ROC-AUC comparison across the three supervised models.

Random Forest achieved the best overall binary detection performance in this archive, with accuracy = 0.934, F1 = 0.966, and ROC-AUC = 0.846. Gradient Boosting matched Random Forest on accuracy and F1 and was only marginally lower in ROC-AUC. The LSTM was slightly weaker in ROC-AUC (0.823) while delivering almost the same accuracy and F1.

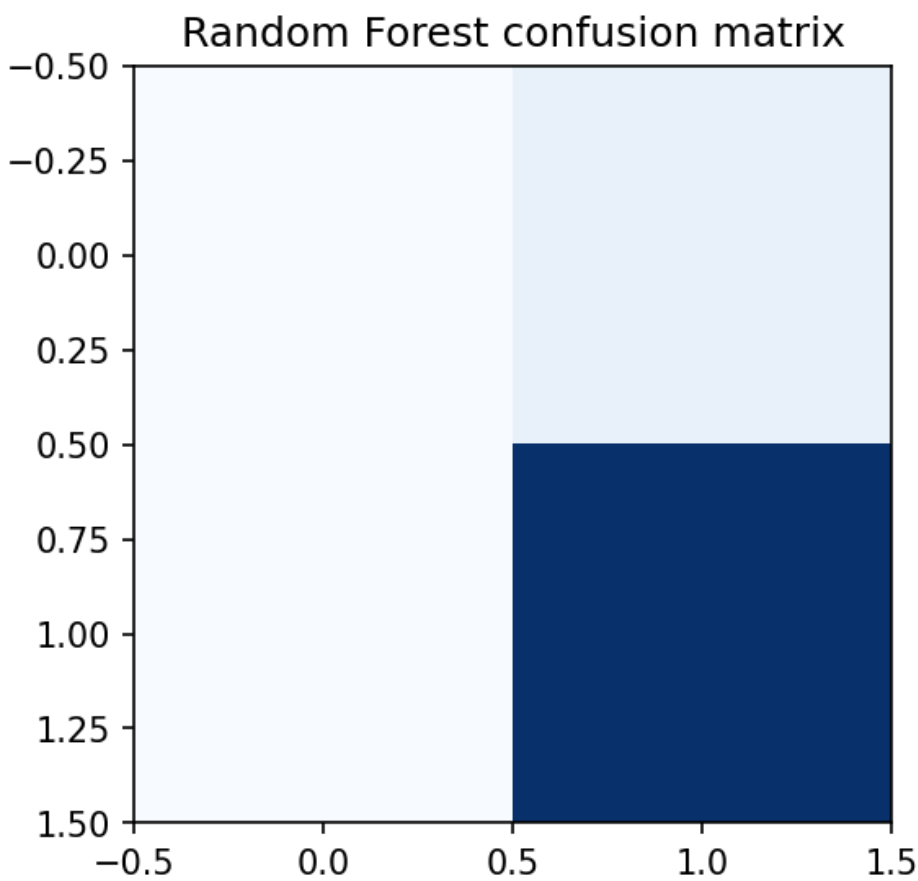


Figure 11. Random Forest confusion matrix.

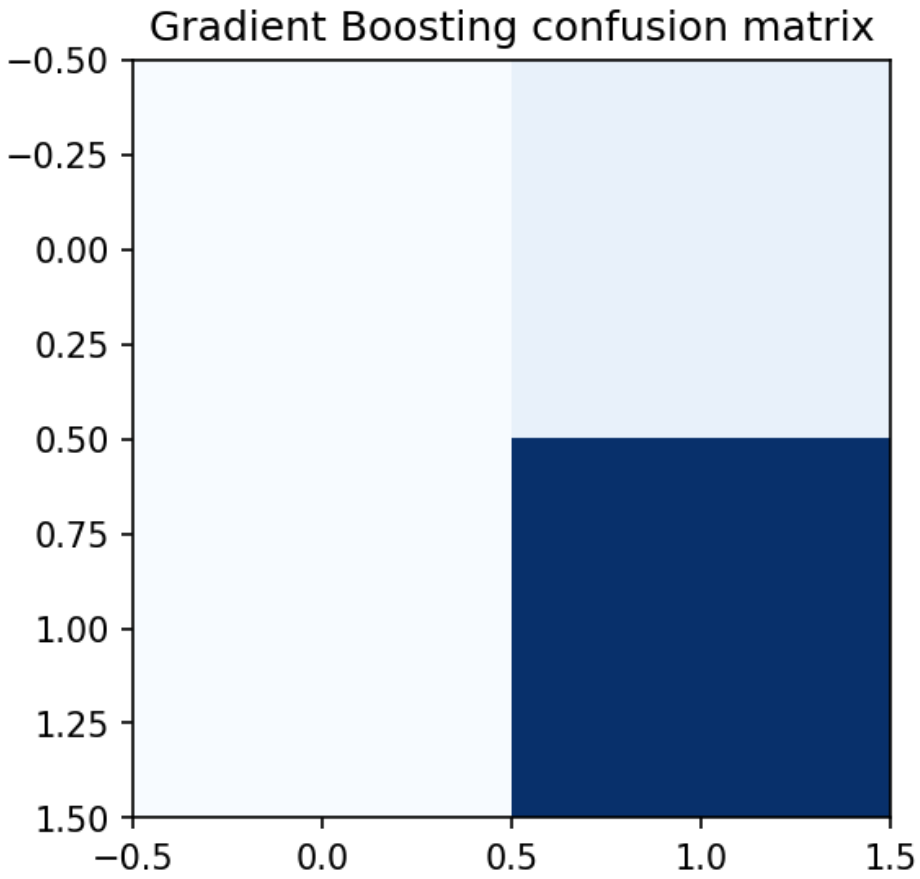


Figure 12. Gradient Boosting confusion matrix.

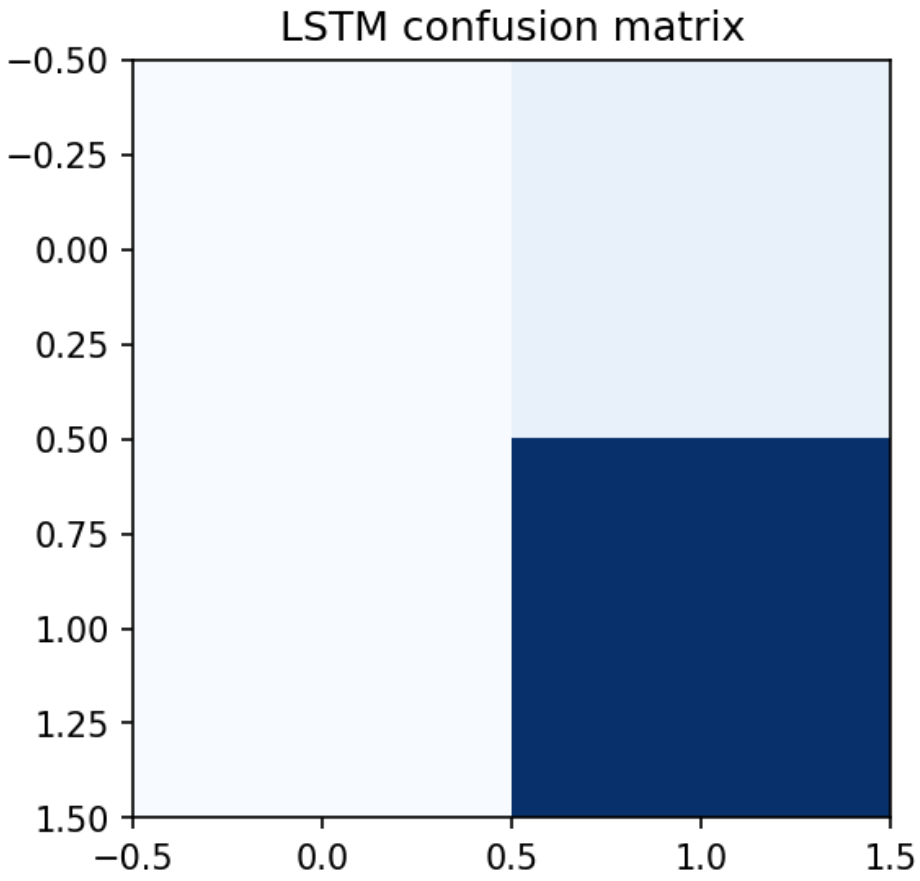


Figure 13. LSTM confusion matrix.

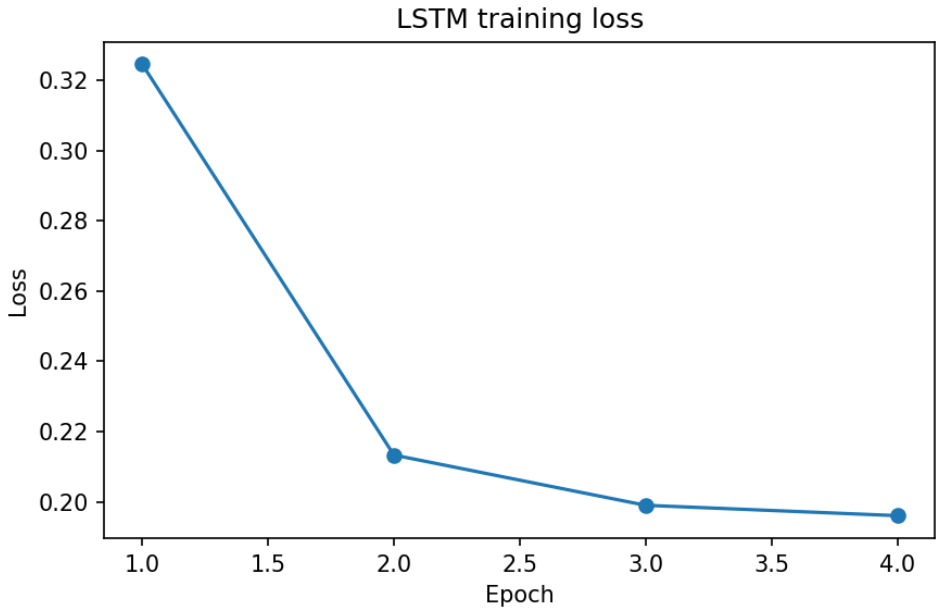


Figure 14. LSTM training loss across epochs.

The near-equal accuracy across all three models should be interpreted carefully because the binary task is highly imbalanced. The ROC-AUC values are more informative here: they show that the tree-based methods separated normal from faulty conditions better than the LSTM on this dataset split.

The LSTM loss curve decreased rapidly from 0.325 to 0.196 within four epochs, which indicates that sequence training was stable and that the model learned quickly from the available trajectories. However, the resulting ROC-AUC suggests that dynamic sequence information did not provide a decisive advantage over strong tabular baselines in this particular harmonized formulation.

Engineering Interpretation in the Acetic-Acid Automation Context

Within the stated industrial framing, these results support a layered process-monitoring strategy for an automated acetic-acid production scheme. First, a lightweight supervised detector such as Random Forest or Gradient Boosting can be used for rapid abnormality screening. Second, unsupervised embeddings and clustering can be used offline by engineers to visualize recurring abnormal operating regimes, identify families of similar upsets, and support root-cause analysis.

The fact that a small set of sensors shows strong mean shifts, while permutation sensitivity remains diffuse, is consistent with a coupled process in which disturbances propagate through multiple units. In such systems, simple thresholding on one signal is rarely enough. Multivariate monitoring is the more robust approach.

From a control-room perspective, the best-value result in this archive is not a single embedding plot but the combination of:

(1) strong binary classifiers for detection,

(2) correlation and fault-pattern heatmaps for process understanding, and

(3) nonlinear clustering views, especially t-SNE/HDBSCAN, for exploratory diagnosis.

Limitations and Recommended Next Steps

Several limitations is when interpreting the archive.

First, the files were harmonized to the common 50-sensor subset (FT1-7,9,12, PT4,7, 10, TT1-4, 8, 9,12-16, LT2-5), so any information contained only in the dropped variables was not used.

Secondly, the unsupervised comparisons were run on a balanced subset rather than the full dataset, which improves stability and interpretability but changes the clustering problem. Benchmark for process automation analytics.

The most useful next extensions would be a multiclass supervised fault classifier, a more rigorous dynamic fault-diagnosis study using longer lag windows and per-fault sequence models, SHAP or partial-dependence analysis for richer interpretability, and a fully native **UMAP/PaCMAP/TriMAP** comparison if package support is available.

Conclusion

Overall, the archived project demonstrates that the harmonized TEP dataset retains enough information for both visual fault exploration and high-performing binary fault detection.

1. Among the unsupervised experimental runs, nonlinear manifold embeddings paired with HDBSCAN were the most informative.
2. Among the supervised runs, Random Forest and Gradient Boosting slightly outperformed LSTM while being simpler to train and explain. The package therefore supports a coherent conclusion: the process data are fault-informative, it also has multivariate monitoring which is necessary, and the strongest practical workflow combines EDA, nonlinear visualization, and tree-based binary detection.